Candidate's name:.....

Signature:.....

P525/2
CHEMISTRY
(THEORY)
Paper 2
August, 2026
3¼ hours

MASAKA DIOCESAN EXAMINATIONS BOARD
JOINT MOCK EXAMINATIONS 2026
Uganda Advanced Certificate of Education
CHEMISTRY
(Practical)
Paper 2
3 hours 15 minutes

INSTRUCTIONS TO CANDIDATES:

This paper consists of **two compulsory** *examination items. Responses to these items are to be written in the spaces provided. Use* **blue** *or* **black** *ink.*

All working **must** *be clearly shown.*

Mathematical tables and silent non-programmable scientific calculators may be used.

You are **not** *allowed to use reference books (i.e. text books, booklets on qualitative analysis etc.)*

You are advised to carefully read the items, make sure you have all the apparatus and chemicals you may need and then **plan** *accordingly before starting.*

ITEM No.	BASIS CODE	LEVEL	SCORERiS INITIALS
1	QP_1		
	QM_1		
	QD_1		
	QC_1		
2	PA_2		
	SA_2		
	CA_2		
	CR_2		

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ITEM ONE:

Scenario:

In food preparation, the purity of substances affects taste, safety, and shelf life. Your S6 chemistry class is investigating why some locally prepared fruit juices spoil quickly while others remain fresh. A nearby juice vendor reports that a recently purchased batch of citric acid, which is her most preferred preservative, is not effective, as some juice becomes too sour while other batches spoil faster than expected.

Your teacher explained that for citric acid to be suitable, its percentage purity should be between 99.0-99.9%. If the purity is below this range, the sample may contain impurities, making it unsuitable for use and requiring rejection or return to the supplier.

Before analysing the sample (Solid J), you found that the available sulphuric acid has an unknown concentration. The teacher has advised to first determine its concentration to ensure accurate results.

Sulphuric acid reacts with sodium hydroxide according to the equation:

$$H_2SO_4(aq) + 2NaOH(aq) \rightarrow Na_2SO_4(aq) + 2H_2O(l)$$

Citric acid reacts with sodium hydroxide according to the equation;

$$C_6H_8O_7(aq) + 3NaOH(aq) \rightarrow Na_3C_6H_5O_7(aq)(aq) + 3H_2O(l)$$

$$(H = 1; C = 12; O = 16)$$

The teacher further guided to first titrate a fixed volume of sodium hydroxide solution (FA2) with sulphuric acid to obtain consistent results. You will then react exactly 1.2g of citric acid with 50 cm¥of sodium hydroxide solution (FA3) which is of course excess for such a reaction, dilute the solution to 250 cm¥ and titrate fixed portions of it with the same sulphuric acid.

Your findings will be used to advise the juice vendor on whether to use or reject the citric acid.

You are provided with the following;

FA1 which is sulphuric acid of unknown concentration.

FA2 which is a 0.1M sodium hydroxide solution.

FA3 which is a 1M solution of sodium hydroxide.

Solid J which is citric acid.

Phenolphthalein indicator

Other laboratory apparatus.

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Task;

As a chemistry learner;

Plan and carry out a scientific investigation and advise the juice vendor accordingly.

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ITEM TWO:

SCENARIO:

The production of pesticides requires careful quality control because an inappropriate nature of the raw materials used can affect the product effectiveness and create safety risks during storage and transportation.

The manufacture of many pesticides requires an organic raw material that contains two functional groups, each contributing to different stages of the production process. If either functional group is absent or has been altered during storage or transportation, the raw material may fail to produce the required pesticide or lead to unwanted by-products.

A pesticide manufacturing company has recently received a consignment of this raw material from a new supplier. Before it is introduced into production, the quality control department has decided that its identity must be verified because the supplier's certificate of analysis was found to be incomplete after delivery.

The quality control manager has submitted a sample of the raw material to your laboratory for analysis. You are provided with **Sample X**, together with the available reagents and apparatus.

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Task

Carry out a systematic qualitative analysis of **Sample X** to identify the nature of the raw material and determine whether the sample is suitable for use in the manufacture of pesticides. Hence, make an appropriate recommendation to the factory management. (***Carry out about 8 tests***)

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